Ordinary Differential Equation

- **Ordinary Differential Equations (ODE)** relate a function $y(t)$ to its derivatives.
- An **Autonomous** ODE has no explicit dependence on t.
- The order of the highest derivative in an ODE is the **Order**.
- A first order **Initial Value Problem (IVP)** is an ODE with an **Initial Condition (IC)** $y(t_0) = y_0$.
- A function satisfying the IC and the ODE for $a < t_0 < b$ solves the IVP on (a, b).

Example: Solution of a First-Order IVP

Solve the IVP $y' = 2y$, $y(0) = 3$.

The solution of the IVP is $\boxed{y(t) = 3e^{2t}}$

Check: $y'(t) = 6e^{2t} = 2y(t)$ and $y(0) = 3$.

Second Order to First Order

1. Start with a second-order IVP

$$y'' = g(t, y, y'), \qquad y(t_0) = a_1, \qquad y'(t_0) = a_2$$

2. Define $y_1 = y$ and $y_2 = y'$. Then

$$\begin{aligned} y_1' &= y_2 & y_1(t_0) &= a_1 \\ y_2' &= g(t, y_1, y_2) & y_2(t_0) &= a_2 \end{aligned}$$

3. With $\boldsymbol{y} = (y_1, y_2)$ and $\boldsymbol{y}_0 = (a_1, a_2)$, write

$$\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y}), \qquad \boldsymbol{y}(t_0) = \boldsymbol{y}_0, \qquad \boldsymbol{f}(t, \boldsymbol{y}) = \begin{pmatrix} y_2 \\ g(t, y_1, y_2) \end{pmatrix}$$

Mass and Spring

Example: Second Order to First Order

- A mass $m > 0$, damping coefficient $c \geq 0$, spring stiffness $k > 0$, and applied force $g(t)$ give

$$my'' + cy' + ky = g(t)$$

- Displacement $y_1 = y$ and velocity $y_2 = y'$ give

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}'}_{\boldsymbol{y}'} = \underbrace{\begin{pmatrix} y_2 \\ \big(g(t) - cy_2 - ky_1\big)/m \end{pmatrix}}_{\boldsymbol{f}(t,\boldsymbol{y})} \iff \boxed{\boldsymbol{y}' = \boldsymbol{f}(t,\boldsymbol{y})}$$

- Initial displacement and velocity supply the two initial values.

Existence and Uniqueness

First-order IVP

If $\boldsymbol{f}$ and its Jacobian $D_{\boldsymbol{y}}\boldsymbol{f}$ (with entries $\partial f_i/\partial y_j$) are continuous near $(t_0, \boldsymbol{y}_0)$, then

$$\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y}), \qquad \boldsymbol{y}(t_0) = \boldsymbol{y}_0$$

has a unique local solution.

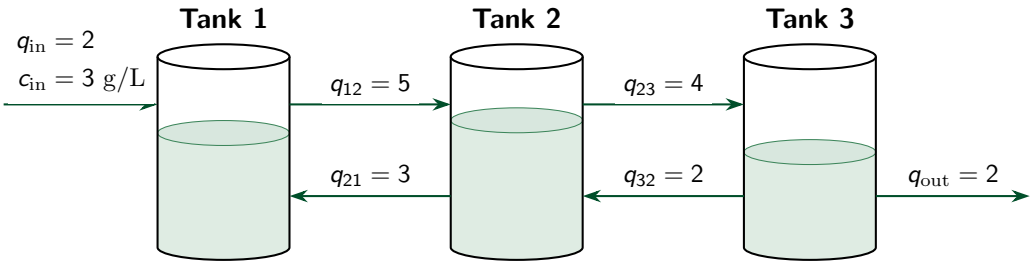
nth-order IVP

If g and its partials wrt $y, y', \ldots, y^{(n-1)}$ are continuous near the IC , then the IVP

$$y^{(n)} = g(t, y, y', \ldots, y^{(n-1)}), \qquad y^{(j)}(t_0) = a_j \quad (j = 0, \ldots, n-1)$$

has a unique local solution.

Three Mixing Tanks



$V_{1,\max} = 100$ L	$V_{2,\max} = 120$ L	$V_{3,\max} = 100$ L
$V_1(0) = 60$ L	$V_2(0) = 80$ L	$V_3(0) = 50$ L
$c_1(0) = 2$ g/L	$c_2(0) = 1$ g/L	$c_3(0) = 0$ g/L

All flow rates are in $\mathrm{L/min}$. Each tank is well mixed.